Positive Square Energy of Every Tricyclic Graph

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Abstract

Let G be a finite simple connected graph with n vertices and $n+2$ edges. We prove $s^+(G) \geq n$, where s^+ is the sum of the squares of the positive adjacency eigenvalues. The proof first splits the cyclic blocks by cyclomatic ranks $1+1+1$, $2+1$, or 3. The first class is the tricyclic cactus theorem. The second consists of one theta block and one cycle block and follows from an exact doubly nonnegative (DNN) sieve and two induced-territory repairs. A rank-three block, after suppressing degree-two paths, has exactly one of four kernels: a four-path theta, a doubled triangle, K_4, or a doubled 4-cycle. We give the correlation dual for the DNN constant, its exact path lemma, and the finite parity certificates and structural repairs covering those kernels. Arbitrary bridges and arbitrary rooted-tree attachments are included. We assert no strict inequality in cases closed only at the DNN boundary.

1 Statement and basic tools

All graphs are finite, simple, and undirected. If $\lambda_1, \ldots, \lambda_n$ are the adjacency eigenvalues, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \qquad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \qquad D(G) = s^+(G) - s^-(G).$$

For a tricyclic graph, $\operatorname{tr} A^2 = 2|E|$ gives

$$s^+(G) + s^-(G) = 2n + 4, \qquad s^+(G) = n + 2 + \frac{1}{2} D(G). \tag{1}$$

Positive and negative square energies were introduced in connection with spectral coloring bounds [1]. The general $n-1$ lower bound was proved by Liu–Tang–Zhang [3]; the edge-surplus conjecture motivating the present statement asks for $s^+(G) \geq n$ when $|E(G)| \geq n+1$ [2].

Theorem 1.1. *Every finite simple connected graph G with $|V(G)| = n$ and $|E(G)| = n+2$ satisfies*

$$\boxed{s^+(G) \geq n}.$$

We use the following pinching consequence of the variational formula for the positive part of a symmetric matrix.

Lemma 1.2 (Induced superadditivity)**.** *If $V(G) = V_1 \dot\cup \cdots \dot\cup V_r$, then*

$$s^+(G) \geq \sum_{j=1}^{r} s^+(G[V_j]).$$

In particular, for $\sigma(H) = s^+(H) - |V(H)|$ one has $\sigma(G) \geq \sum_j \sigma(G[V_j])$.

Proof. For a real symmetric matrix A, $\operatorname{tr}(A_+^2) = \max_{X\succeq 0}(2\operatorname{tr} AX - \operatorname{tr} X^2)$. Insert the block diagonal matrix whose blocks are the positive parts of the principal matrices $A[V_j]$. All cross-block entries of A disappear from the trace, giving the assertion. □

A nonempty connected tree T is bipartite, so its nonzero spectrum is symmetric and $s^+(T) = |E(T)| = |V(T)| - 1$; hence $\sigma(T) = -1$. We shall also use the following attached-bicyclic credits.

Lemma 1.3 (Bicyclic credits)**.** *Let H be connected and bicyclic, allowing rooted trees at every core vertex.*

1. *If H is bipartite, then $\sigma(H) = 1$.*
2. *If its cyclic core is either a two-triangle cactus or $\Theta(1,2,2)$, then $\sigma(H) > 1$.*
3. *Every bicyclic cactus has $\sigma(H) \geq 0$; every nonunicyclic cactus has $s^+(H) > |V(H)|$.*

Proof. The first statement follows from spectral symmetry and $|E(H)| = |V(H)| + 1$. For (2), orient every attached tree toward the core. Its matching recurrence replaces the activity t at each core vertex by

$$Q_{u\to v} = t + \sum_{w \text{ child of } u} Q_{w\to u}^{-1} > 0, \qquad a_v = t + \sum_{u \text{ off core at } v} Q_{u\to v}^{-1} \geq t, \tag{2}$$

and extracts one common positive factor. In the normalized Sachs expansion $i^{-|V|}\det(itI - A)$, every singleton triangle contributes a strictly negative imaginary term. The empty term and the even-cycle or disjoint-pair terms are real. Thus the determinant lies in the open lower half-plane for $t > 0$. Its continuous argument is negative, and the signed Coulson identity

$$D(H) = -\frac{4}{\pi}\int_0^\infty t\, \operatorname{Arg}\big(i^{-|V(H)|}\det(itI - A(H))\big)\, dt \tag{3}$$

gives $D(H) > 0$. Since H is bicyclic, $\sigma(H) = 1 + D(H)/2 > 1$. Statement (3) is the sharp bicyclic-cactus and nonunicyclic-cactus theorem; its proof uses the same tree elimination, the block-additive DNN constant below, and the residual cycle packets [4, 5]. We cite it only in its displayed form. □

2 The DNN dual and the exact path lemma

For an edge-containing graph H, define

$$\kappa(H) = \sup_{\substack{M\succeq 0,\ M\geq 0\\ \mathbf{1}^T M\mathbf{1}>0}} \frac{4\big(\sum_{uv\in E(H)}\sqrt{M_{uv}}\big)^2}{\mathbf{1}^T M\mathbf{1}}. \tag{4}$$

Lemma 2.1 (DNN spectral inequality and correlation dual)**.** *For every graph H,*

$$s^-(H) \leq \kappa(H), \qquad \kappa(H) = \min_{\substack{R\succeq 0\\ R_{vv}=1}} \sum_{uv\in E(H)} \frac{2}{1 - R_{uv}}, \tag{5}$$

where boundary values are interpreted by limits. Moreover κ is additive under one-vertex sums and $\kappa(T) = |E(T)|$ for a tree.

Proof. Write $A = P - B$ for its positive and negative spectral parts. Then $M = B \circ B$ is positive semidefinite and entrywise nonnegative. Also $\mathbf{1}^T M \mathbf{1} = \operatorname{tr} B^2 = s^-$ and $s^-/2 = -\sum_{uv \in E} B_{uv} \le \sum_E \sqrt{M_{uv}}$. Equation (4) yields $(s^-)^2 \le \kappa s^-$.

For weak duality, weighted Cauchy–Schwarz and a correlation matrix R give

$$\mathbf{1}^T M \mathbf{1} - \sum_{uv \in E} 2(1 - R_{uv}) M_{uv} = \langle R, M \rangle + 2 \sum_{uv \notin E} (1 - R_{uv}) M_{uv} \ge 0.$$

Optimizing the weights proves the "$\le$" part of the dual. Conversely, normalize $\langle J, M \rangle = 1$ and use $(\sum_e \sqrt{x_e})^2 = \min_{a_e > 0, \sum a_e = 1} \sum_e x_e / a_e$. Sion minimax and $(\mathcal{S}_+ \cap \mathcal{N})^* = \mathcal{S}_+ + \mathcal{N}$ produce a correlation matrix attaining the reverse inequality; an ε truncation gives boundary points. This proves (5).

For a one-vertex sum, fold nonedge entries into diagonal terms by adding $M_{uv}(e_u - e_v)(e_u - e_v)^T$, then split the Gram vector at the cut vertex orthogonally between the two sides. Both the edge numerator and denominator split, and Cauchy–Schwarz proves subadditivity. Orthogonally gluing near-optimizers with the optimal relative scale proves the reverse inequality. Iterating $\kappa(K_2) = 1$ gives the tree formula. □

Lemma 2.2 (Exact path elimination)**.** *Suppose a kernel edge uv is replaced by a path of length ℓ. If the endpoint correlation is $r = R_{uv}$, its exact excess over the baseline ℓ in the dual is*

$$f_\ell(r) = \ell \tan^2\left(\frac{\arccos((-1)^\ell r)}{2\ell}\right). \tag{6}$$

For a fixed r and parity, $f_{\ell+2}(r) < f_\ell(r)$ unless the endpoint angle vanishes. Different paths may use the same endpoint vectors, so the minimum of the sum of these path costs over kernel correlation matrices is exactly $\kappa - L$, where L is the sum of the path lengths.

Proof. Alternately negate the unit Gram vectors along the path. The transformed endpoint angle is $\beta = \arccos((-1)^\ell r)$. The spherical triangle inequality says the sum of the ℓ successive angles is at least β; convexity of $\sec^2(x/2)$ makes equal angles optimal. Equally spaced planar vectors on the shortest arc attain the bound, proving exactness. With $h_s(x) = s \tan^2(x/s)$ and $z = x/s$,

$$\frac{d}{ds} h_s(x) = \tan z \,(\tan z - 2z \sec^2 z) < 0$$

because $\sin z \cos z < z$. This proves parity monotonicity. □

If a rank-three core has L edges, it has $L - 2$ vertices. Thus

$$\kappa \le L + 2 \quad \Longrightarrow \quad s^+ = 2L - s^- \ge L - 2 = |V|. \tag{7}$$

By Lemma 2.1, the same implication survives arbitrary rooted-tree attachments: each attached tree edge adds one to both κ and the vertex count.

3 Cyclic-block classification

We use the standard block decomposition: a block is a maximal connected subgraph with no cut vertex, with bridges regarded as two-vertex blocks. Thus a cycle meeting the rest of the graph in a single vertex is a separate rank-one block. All degrees used when suppressing a cyclic block are degrees within that block.

The cyclomatic rank of a connected graph is $|E| - |V| + 1$. Bridge blocks have rank zero, and ranks add over the nontrivial blocks.

Proposition 3.1 (Block alternatives)**.** *The positive ranks of the cyclic blocks of a connected tricyclic graph are exactly one of*

$$1+1+1, \qquad 2+1, \qquad 3.$$

Rank-one blocks are cycles. A simple rank-two 2-connected block is a theta. A rank-three 2-connected block, after suppressing each maximal path whose internal vertices have degree two in the block, has exactly one of the following loopless multigraph kernels:

1. *four parallel edges between two branch vertices;*
2. *a triangle with two adjacent sides doubled;*
3. *K_4;*
4. *a 4-cycle with two opposite sides doubled.*

Every omitted part is a bridge path or a rooted tree attached at one vertex.

Proof. Block ranks are positive integers summing to three, giving the three partitions. A rank-one 2-connected graph is a cycle. In rank two, $\sum_v (d(v)-2) = 2$ after suppressing degree two; 2-connectivity forces two degree-three vertices joined by three parallel kernel edges.

Suppression cannot create a loop. Such a loop would come from a cycle based at a branch vertex u whose other vertices all have degree two in the block. If the block had any part outside that cycle, deleting u would separate the cycle interior from that part, contrary to 2-connectivity. If it had no outside part, the block would itself be a cycle and have rank one.

In rank three the same sum is four. Hence the branch degrees are either $(4,4)$, $(4,3,3)$, or $(3,3,3,3)$. The first gives four parallel edges. For $(4,3,3)$, looplessness and 2-connectivity force the degree-four vertex to have two edges to each degree-three vertex, with one edge between the latter: the doubled triangle. A cubic loopless 2-connected multigraph on four vertices is either simple, hence K_4, or has parallel edges. Choose a parallel pair uv. It cannot be a triple edge, since then $\{u,v\}$ would be a component, so it is exactly double. Each of u, v has one remaining incidence. If both met the same remaining vertex x, the fourth vertex y could meet only x; making $d(y) = 3$ would already give three xy edges and force $d(x) \geq 5$. Thus the two incidences meet the two remaining vertices x, y separately. Now x and y each have residual degree two, and their only available mutual edge therefore occurs twice. Consequently the kernel has the double edges uv, xy and two single connectors, hence is a 4-cycle with opposite sides doubled. Simplicity of the original graph says at most one path in each parallel family may have length one. Reversing suppression and leaf deletion gives the final assertion. □

4 The $1+1+1$ and $2+1$ blocks

Proposition 4.1. *Every tricyclic graph with block ranks $1+1+1$ satisfies $s^+ > n$.*

Proof. Such a graph is a cactus of cyclomatic rank three, including all bridge paths and tree attachments. Apply Lemma 1.3(3). □

For a theta with path lengths a, b, c, put $L = a+b+c$ and

$$\Delta(a,b,c) = \min_{0\leq\theta\leq\pi} \left(\sum_{\ell \text{ even}} \ell \tan^2 \frac{\theta}{2\ell} + \sum_{\ell \text{ odd}} \ell \tan^2 \frac{\pi-\theta}{2\ell} \right). \tag{8}$$

For a cycle put $\epsilon_q = 0$ for even q and $\epsilon_q = q\tan^2(\pi/(2q))$ for odd q.

Lemma 4.2 (Theta–cycle DNN sieve)**.** *For a graph whose cyclic blocks are $\Theta(a,b,c)$ and C_q,*

$$s^+(G) \geq n + 2 - \Delta(a,b,c) - \epsilon_q.$$

This is at least n except possibly for

$$\Theta(1,2,r) + C_3 \quad (r \geq 2), \qquad \Theta(1,2,2) + C_5.$$

Proof. Lemma 2.2 gives $\kappa(\Theta) = L + \Delta$ and $\kappa(C_q) = q + \epsilon_q$; one-vertex additivity includes every intervening bridge and attached tree. The displayed spectral bound follows from Lemma 2.1. We now give the numerical classification needed for the sieve. Put $x = \theta/2$ and

$$h_s(y) = s \tan^2(y/s).$$

For fixed $0 < y < \pi/2$ this is strictly decreasing in s, since, with $z = y/s$,

$$\frac{d}{ds} h_s(y) = \frac{\tan z}{\cos^2 z}\big(\sin z \cos z - 2z\big) < 0. \tag{9}$$

Thus, within a fixed parity, replacing an even length by 2 or an odd length by 1 or 3 can only increase the objective in (8). Because the graph is simple, at most one of the three theta paths has length one.

First suppose that no path has length one, and let e be the number of even paths. Parity monotonicity bounds the objective from above by

$$F_e(x) = 2e \tan^2(x/2) + 3(3-e) \tan^2\big((\pi/2 - x)/3\big).$$

For $e = 0$ and $e = 3$, take respectively $x = \pi/2$ and $x = 0$, obtaining zero. For the two mixed cases the explicit tests

$$F_1(\pi/4) < 1, \qquad F_2(\pi/6) < 1$$

follow from $\tan(\pi/8) = \sqrt{2} - 1$, $\tan(\pi/12) = 2 - \sqrt{3}$, and $\tan(\pi/9) < \tan(\pi/8)$; after substitution the two upper bounds are $48 - 4\sqrt{2} - 24\sqrt{3}$ and $37 - 6\sqrt{2} - 16\sqrt{3}$, both less than one by successive squaring of positive sides. Hence $\Delta < 1$ whenever there is no length-one path.

Now suppose that there is one such path. Its contribution is $\cot^2 x$. If the other two paths are odd, the choice $x = \pi/2$ gives $\Delta = 0$. If exactly one of them is even and its length is at least 4, monotonicity and the test $x = \pi/3$ give

$$\Delta \leq \frac{1}{3} + 4 \tan^2 \frac{\pi}{12} + 3 \tan^2 \frac{\pi}{18} < 1. \tag{10}$$

Indeed $\tan(\pi/18) < \tan(\pi/12) < 1/3$, and the resulting inequality follows from $48\sqrt{3} > 83$, whose square is $6912 > 6889$. If both remaining paths are even and both have length at least 4, the same test gives

$$\Delta \leq \frac{1}{3} + 8 \tan^2 \frac{\pi}{12} < 1, \tag{11}$$

because $12(2 - \sqrt{3})^2 < 1$, equivalently $6889 < 6912$ after squaring.

It remains precisely to consider sorted triples $(1,2,r)$, $r \geq 2$. The first two path terms satisfy, on writing $u = \tan(x/2)$,

$$\cot^2 x + 2 \tan^2(x/2) = \frac{1}{4u^2} - \frac{1}{2} + \frac{9}{4} u^2 \geq 1. \tag{12}$$

The third term is positive in the interior; at whichever endpoint it can vanish, the expression in (12) is either divergent or equal to 2. Compactness therefore makes the minimum strictly larger than 1. This proves the exact classification

$$\Delta(a,b,c) > 1 \quad \Longleftrightarrow \quad (a,b,c) = (1,2,r) \text{ after sorting}, \quad r \geq 2. \tag{13}$$

For $r = 2$, the whole objective is

$$\frac{1}{4u^2} - \frac{1}{2} + \frac{17}{4}u^2,$$

so minimization at $u^2 = 1/\sqrt{17}$ gives the exact maximum

$$\Delta(1,2,2) = \frac{\sqrt{17}-1}{2}. \tag{14}$$

For even $r \geq 4$, (9) makes the objective pointwise smaller than for $r = 2$. Moreover, away from $r = 2$ one has the uniform strict bound $\Delta(1,2,r) < 4/3$: for odd $r \geq 3$, test $x = \pi/3$ and use $\tan(\pi/18) < 1/3$, while for even $r \geq 4$ the same test reduces to $1 + 4(2-\sqrt{3})^2 < 4/3$, already contained in $6889 < 6912$. Since $4/3 < (\sqrt{17}-1)/2$, this also proves that (14) is the global maximum.

Finally, the odd-cycle excess is $h_q(\pi/2)$, so (9) says that it decreases strictly with odd q. The first two values and a convenient bound for the rest are

$$\epsilon_3 = 1, \qquad \epsilon_5 = 5 - 2\sqrt{5} < \frac{2}{3}, \qquad \epsilon_q \leq \epsilon_7 < \frac{2}{5} \quad (q \geq 7 \text{ odd}). \tag{15}$$

Here the first two identities are exact. For the last inequality, use $\sin t < t$, $\cos t > 1 - t^2/2$, and $\pi < 22/7$ at $t = \pi/14$; clearing denominators gives $7\tan^2 t < 2/5$. Even cycles have zero excess.

We need $\Delta + \epsilon_q \leq 2$. For $q = 3$, (13) leaves exactly every $(1,2,r)$. For $q = 5$, the bound $\Delta < 4/3$ disposes of all those triples except $(1,2,2)$, and (14) shows that this remaining triple really exceeds the DNN threshold: $(\sqrt{17}-1)/2 + 5 - 2\sqrt{5} > 2$, since $\sqrt{17} + 7 > 4\sqrt{5}$. For odd $q \geq 7$, $(\sqrt{17}-1)/2 < 8/5$ and (15) give a strict total below 2; even q is immediate. Thus the exact residuals of this DNN sieve are the two rows stated. They are failures of the DNN threshold, not graph counterexamples. □

Proposition 4.3. *Every tricyclic graph with block ranks* $2+1$ *satisfies* $s^+(G) > n$.

Proof. Only the rows of Lemma 4.2 need repair. In $\Theta(1,2,r) + C_3$, choose an internal theta vertex not used by the unique block-tree route to the external triangle: for $r = 2$ choose one of the two length-two internal vertices; for $r \geq 3$ choose one of two internal vertices of the r-path. Delete that vertex together with its rooted branch. The deleted induced territory is a nonempty tree of credit -1, while the connected complement is an attached two-triangle bicyclic cactus of credit > 1 by Lemma 1.3. Lemma 1.2 gives positive total credit.

For $\Theta(1,2,2) + C_5$, a separating bridge gives two induced territories. The theta territory has credit > 1 and the attached pentagon has credit at least $-(\sqrt{5}-2)$, by the one-cycle phase comparison $D \geq -2(\sec(\pi/5) - 1)$. Their sum is positive. If the blocks share a cut vertex, keep the theta and that cut vertex in one territory and put the other four pentagon vertices, with all their rooted branches, in the other. The latter is a nonempty tree of credit -1 and the former has credit > 1. These assignments cover every bridge length, cut location, and rooted branch. □

5 The four rank-three kernels

5.1 Four-path theta

Proposition 5.1. *Every simple subdivision of the four-parallel-edge kernel, with arbitrary rooted trees, satisfies $s^+ \geq n$.*

Proof. For path lengths $\ell_1, \ldots, \ell_4$, Lemma 2.2 reduces the excess to

$$\Phi(\theta) = \sum_{\ell \text{ even}} \ell \tan^2 \frac{\theta}{2\ell} + \sum_{\ell \text{ odd}} \ell \tan^2 \frac{\pi - \theta}{2\ell}.$$

If no length is one, monotonicity reduces even lengths to two and odd lengths to three. Writing $x = \theta/2$ and letting e be the number of even paths, test $2e \tan^2(x/2) + 3(4-e)\tan^2((\pi/2 - x)/3)$ at an endpoint for $e = 0, 1, 3, 4$; for $e = 2$ its right derivative at $x = 0$ is negative. The minimum is at most two.

If one length is one (there cannot be two by simplicity), its term is $\cot^2 x$. For zero, one, or two even paths among the remaining three, $x = \pi/2$ or $x = \pi/3$ gives a value below two. With three even paths,

$$\Phi(2x) \leq \cot^2 x + 6 \tan^2(x/2).$$

At $u = \tan(x/2)$ the right side is $1/(4u^2) - 1/2 + 25u^2/4$, with minimum two at $u^2 = 1/5$. Strict path monotonicity improves this unless the lengths are $(1, 2, 2, 2)$, where direct substitution gives equality. Hence $\kappa \leq L + 2$, and (7) proves the claim. Notice that the exceptional core has spectrum $\{3, -2, -1, 0, 0\}$ and actually has $s^+ = 9 > 5$; we do not infer a general strict statement from this isolated computation. □

5.2 Finite kernel certificates

The remaining arguments repeatedly use the following finite-certificate principle. Encode a path parity by 0 for even and 1 for odd. Switching at a branch vertex toggles all incident bits. For fixed physical parities, Lemma 2.2 reduces all lengths to the first simple lengths: two for even, one for odd, except that two parallel odd paths use $1, 3$. A displayed correlation Gram matrix with path excess at most two then proves every longer realization of that physical row. A switching label is used only to organize rows; it is never used to transport a certificate between physical rows with different first lengths.

Lemma 5.2 (Doubled 4-cycle certificate)**.** *Every simple subdivision of the doubled-4-cycle kernel satisfies $s^+ \geq n$. The proof establishes $\kappa \leq L + 2$ in the DNN rows and handles the remaining rows by structural repair.*

Proof. Write the paths as $a, A : 01$, $b : 12$, $c, C : 23$, $d : 30$. There are eight switching invariants and 64 realizable labelled parity words. Table 1 records the old 36-word slice having no odd–odd doubled pair. Exact planar Gram vectors $u_j = (\cos(\pi k_j/6), \sin(\pi k_j/6))$ give excess at most two in 28 of these rows. The two boundary certificate types have contributions, in a suitable order, $2/3, 1/3, 0, 2/3, 1/3, 0$, summing to two; all other listed certificates have strict slack. This finite table is proved by direct substitution in (6) in [6].

The missing 28 words have at least one odd–odd doubled side. Normalize one such side to first-simple lengths $(1, 3)$. The other doubled side has type $EE = (2, 2)$, $EO = (1, 2)$, or $OO = (1, 3)$,

invariant	first simple row	physical rows	disposition
000	000000	2	DNN
001	000001	2	DNN
010	000010	4	DNN
011	000011	4	DNN
100	010000	4	DNN
101	010001	4	DNN
110	010010	8	four DNN, four DNN or deletion
111	010011	8	four DNN, four DNN or deletion

Table 1: Doubled-4-cycle no-odd–odd slice; the counts total 36 of the 64 realizable labelled parity words.

and the two connectors have independent parities, leaving twelve normalized states. The complete planar $\pi/3$-angle ledger in [6] has exact costs among

$$0, \quad 2\beta + 2\alpha, \quad 4\alpha, \quad 4\beta, \quad 4\beta + \alpha, \quad 3\beta + \alpha, \quad 2\beta + \alpha + \gamma,$$

where $\alpha = 2\tan^2(\pi/12)$, $\beta = \tan^2(\pi/6)$, and $\gamma = 3\tan^2(\pi/18)$. Exact radical and triple-angle comparisons give $\alpha < 1/6$, $\beta < 17/50$, and $\gamma < 1/10$; the finite checker also reconstructs the terms by rational Taylor enclosures. The largest resulting upper bound is $4(17/50) + 1/6 = 229/150 < 2$. Fixed-physical-parity monotonicity therefore covers arbitrary longer lengths in all 28 formerly omitted words, and one-vertex additivity covers every rooted-tree attachment.

For completeness, the eight rows requiring repair in the old slice form one orbit under kernel automorphisms and interchange of parallel-path names; in the fixed switching normalization, four lie in class 110 and four in class 111. Normalize this orbit so each doubled side has one even and one odd path and the connectors have opposite parity. If a member of a doubled pair is longer than canonical, the exact planar certificates at lengths $(4, 1, 1, 2, 1, 2)$ and $(2, 3, 1, 2, 1, 2)$ have respective excesses bounded by

$$4\tan^2\frac{3\pi}{32} + 3\tan^2\frac{\pi}{8} + 2\tan^2\frac{\pi}{16} + 2\tan^2\frac{3\pi}{16} < 1.862,$$

$$2\tan^2\frac{\pi}{16} + 4\tan^2\frac{\pi}{8} + 2\tan^2\frac{3\pi}{16} < 1.662.$$

If both doubled pairs are canonical, open an internal vertex of the even connector. The deleted territory is a tree of credit -1; the complement is a two-triangle cactus of credit > 1. Together with the odd–odd ledger this exhausts all 64 parity words and includes all attachments. □

Lemma 5.3 (Doubled-triangle certificate)**.** *Every simple subdivision of the doubled-triangle kernel, with arbitrary rooted trees, satisfies $s^+ \geq n$.*

Proof. Write $a, A : 01$, $b, B : 02$, and $c : 12$. A doubled pair has first-simple type $EE = (2, 2)$, $EO = (1, 2)$, or $OO = (1, 3)$. Table 2 is the complete 32-row physical census. In eleven orbits an exact rational 3×3 correlation matrix gives excess at most two. The entries are obtained from multiple-angle formulas; for example, with parameter t,

$$r_{EE} = r_{EO} = \frac{1 - 6t^2 + t^4}{(1 + t^2)^2}, \quad e_{EE} = 4t^2, \quad e_{EO} = \left(\frac{1 - t^2}{2t}\right)^2 + 2t^2,$$

pair types	rows for even c	rows for odd c	DNN rows	structural rows
EE, EE	1	1	2	0
EE, EO	4	4	8	0
EE, OO	2	2	4	0
EO, EO	4	4	4	4
EO, OO	4	4	8	0
OO, OO	1	1	2	0
total	16	16	28	4

Table 2: Doubled-triangle physical parity census.

and, for $q = (3t - t^3)/(1 - 3t^2)$, $r_{OO} = -(1 - q^2)/(1 + q^2)$ and $e_{OO} = q^2 + 3t^2$. Substitution, nonnegative determinant, and the exact excess fractions are listed in [7]; thus no floating optimization is used.

The four structural rows are EO, EO with odd c. If any parallel path is noncanonical, exact planar certificates reduce to $(4, 1, 2, 1, 1)$ or $(2, 3, 2, 1, 1)$ and have excess below $229/120$ or $31/20$. Otherwise both parallel pairs are $\{1, 2\}$. If $c \geq 3$, open c and leave a two-triangle cactus; if $c = 1$, open the even member of one pair and leave $\Theta(1, 2, 2)$. Lemmas 1.2 and 1.3 pay the deleted tree. These alternatives exhaust every length and attachment. □

Lemma 5.4 (K_4-subdivision certificate)**.** *Every simple subdivision of K_4, with arbitrary rooted trees, satisfies $s^+ \geq n$.*

Proof. Switching the six path parities has eight classes, each containing eight physical rows. Label the branch vertices $1, 2, 3, 4$, switch so that the three paths to vertex 4 are odd, and let $\epsilon_{12}, \epsilon_{13}, \epsilon_{23}$ indicate which normalized triangle paths are even. The word 000 is the all-odd class. For a word of weight one use, up to a permutation of $1, 2, 3$, the normalized planar angles $(2\pi/3, 4\pi/3, 4\pi/3, 0)$; for weight two use $(2\pi/3, 4\pi/3, \pi, 0)$; for weight three use $(\pi, \pi, \pi, 0)$. After undoing the switch on the vectors, but not on the physical path lengths, every canonical physical path has cost 0, $a = 2\tan^2(\pi/12) = 14 - 8\sqrt{3}$, or $b = \tan^2(\pi/6) = 1/3$. The complete eight-switch substitution for either a weight-one or a weight-two class is Table 3; permutations give all six such classes. The weight-three template has zero cost in all eight rows. This is the exact physical-row table of [9], not a transport of canonical lengths across a switch.

number of rows	number of a terms	number of b terms	total excess
1	0	5	$5/3$
2	2	3	$2a + 1$
4	3	2	$3a + 2/3$
1	4	1	$4a + 1/3$

Table 3: Exact eight-switch physical-cost census for each weight-one or weight-two K_4 class. There are three classes of each weight; the remaining weight-three class has eight zero-cost rows. Thus the table and that last class account for all $6 \cdot 8 + 8 = 56$ non-all-odd physical rows.

Exactly, $a < 1/6$ because $\sqrt{3} > 83/48$, whose square is the integer inequality $6912 > 6889$. Hence the four totals in Table 3 are at most $5/3 < 2$. Fixed-physical-parity path monotonicity proves all longer realizations, and one-vertex additivity includes all rooted trees. In particular, the seven-class ledger proves the strict bound $\kappa < L + 2$ in all 56 rows. The normalized templates, all

eight switch triples, the four-type physical substitution table, and these exact inequalities are given explicitly in [9].

Here is the independent all-length cover of the last switching class [8]. Its physical parities have the form $p_{ij} = 1 \oplus s_i \oplus s_j$. After replacing branch vector u_i by $(-1)^{s_i} u_i$, every transformed endpoint angle is the corresponding all-odd angle, while the coefficient and denominator remain the physical path length. Call a physical path unit if its length is one and long otherwise. Every even physical path is therefore long. A nonconstant switch has exactly three or four even paths, the edges of a nontrivial cut of K_4.

With at least three long paths, first use four regular-simplex vectors, $R_{ij} = -1/3$. A unit path contributes $1/2$, a long odd path contributes less than $1/6$, and a long even path contributes less than $1/4$. Thus the physically all-odd row with at least three long paths and every switched row with at least four long paths cost less than two. The only row left by this estimate is a three-edge even cut and the complementary odd unit triangle. Relabel the cut as the star from vertex 0 and use the physical correlation matrix

$$R = \begin{pmatrix} 1 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1 & -11/24 & -11/24 \\ 1/6 & -11/24 & 1 & -11/24 \\ 1/6 & -11/24 & -11/24 & 1 \end{pmatrix}.$$

Its eigenvalues are $35/24, 35/24, 13/12, 0$, so it is positive semidefinite. The three even length-two cut paths and three odd unit triangle paths have exact total excess

$$3\frac{38 - 8\sqrt{21}}{5} + 3\frac{13}{35} = \frac{837 - 168\sqrt{21}}{35} < 2;$$

the last inequality follows from $168^2 \cdot 21 - 767^2 = 4415 > 0$. Fixed-physical-parity path monotonicity retains this Gram for every longer realization of the row.

If exactly two paths are long, the switch must be constant, so all paths are physically odd. There are two placements. Opposite paths use branch angles $(0, \pi/4, \pi, 5\pi/4)$ and give $8\tan^2(\pi/8) = 24 - 16\sqrt{2} < 2$. Adjacent paths use $(0, 3\pi/8, 13\pi/8, \pi)$ and give

$$6\tan^2\frac{5\pi}{48} + \tan^2\frac{\pi}{8} + 2\tan^2\frac{3\pi}{16} < \frac{71}{40} < 2.$$

Exactly one long path also forces the switch to be constant. Hence all paths are odd, the unique long path has length at least three, and the other five are units. Open the long path at an internal vertex. The complement is an attached $\Theta(1,2,2)$ with strict credit greater than one by Lemma 1.3; the deleted nonempty tree has credit minus one. With no long path every switch is constant and the core is K_4. We include the attached-K_4 argument rather than infer it from the bare spectrum. For any graph H obtained by attaching rooted trees at the four vertices of K_4, put $Z_F(t) = \sum_j m_j(F) t^{|V(F)|-2j} > 0$, where $m_j(F)$ counts the j-edge matchings of F. Grouping Sachs' expansion by its vertex-disjoint cycle collection gives

$$\Psi_H(t) := i^{-|V(H)|} \det(itI - A(H)) = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-2i^{-|C|}) Z_{H - V(\mathcal{C})}(t).$$

Every cycle lies in the K_4, so no two are vertex-disjoint. The empty term and the 4-cycle terms are real, whereas each of the four triangles has factor $-2i$. Therefore

$$\operatorname{Im} \Psi_H(t) = -2 \sum_{T \subseteq K_4,\ T \cong K_3} Z_{H - V(T)}(t) < 0 \qquad (t > 0).$$

Moreover $\Psi_H(t) = \prod_j (t + i\lambda_j)$, so $\sum_j \arctan(\lambda_j/t)$ is its continuous argument tending to zero at infinity. Since the curve remains in the open lower half-plane, that argument is the principal one and is negative. The signed Coulson identity (3) yields $D(H) > 0$. As H is tricyclic, (1) gives $s^+(H) > |V(H)| + 2$, in particular the required claim. This is the attached-K_4 lower-half-plane Sachs packet from [10]; bare K_4 has spectrum $\{3, -1, -1, -1\}$ and $s^+ = 9$. Thus all eight switching classes, including all eight physical rows in the last class, are covered. □

Proof of Theorem 1.1. Apply Proposition 3.1. Block ranks $1+1+1$ are covered by Proposition 4.1, and ranks $2 + 1$ by Proposition 4.3. For one rank-three block, the suppressed kernel is one of the four listed in Proposition 3.1; apply Proposition 5.1 or Lemmas 5.2, 5.3, and 5.4. The DNN additivity and each structural deletion explicitly include every bridge and rooted-tree attachment. The alternatives are exhaustive, proving $s^+(G) \geq n$. □

A Complete doubled-C_4 Gram ledger

This appendix records the old no-odd–odd part of Lemma 5.2; the additional twelve-state odd–odd ledger is stated in the lemma and given in full in [6]. The path order is (a, A, b, c, C, d) and a parity word uses zero for even and one for odd. In Table 4, ℓ is the physical length vector on this restricted slice (replace zero by two and one by one), and $k = (k_1, k_2, k_3)$ means that the four branch vectors have arguments $\pi(0, k_1, k_2, k_3)/6$. Thus

$$R_{ij} = \cos \frac{\pi(k_i - k_j)}{6}.$$

For a path joining i to j, let $q \in \{0, \ldots, 6\}$ be the principal integer satisfying

$$\arccos((-1)^\ell R_{ij}) = \frac{\pi q}{6}.$$

The six entries in the q column follow the path order, and the certified excess is therefore reconstructed, with no numerical optimization, as

$$E = \sum_{r=1}^{6} \ell_r \tan^2 \frac{\pi q_r}{12\ell_r}. \tag{16}$$

Every displayed Gram matrix is planar and hence positive semidefinite with unit diagonal. The rational number in the last column is a strict upper bound unless equality is printed.

Table 4: The 28-row no-odd–odd doubled-C_4 DNN Gram ledger.

parity row	canonical ℓ	(k_1, k_2, k_3)	$(q_a, q_A, q_b, q_c, q_C, q_d)$	exact bound for E
000000	222222	$(0, 0, 0)$	$(0, 0, 0, 0, 0, 0)$	$E = 0$
001001	221221	$(0, 6, 6)$	$(0, 0, 0, 0, 0, 0)$	$E = 0$
000001	222221	$(11, 8, 7)$	$(1, 1, 3, 1, 1, 1)$	$E < 3/5$
001000	221222	$(1, 8, 9)$	$(1, 1, 1, 1, 1, 3)$	$E < 3/5$
000010	222212	$(1, 3, 11)$	$(1, 1, 2, 4, 2, 1)$	$E < 5/4$
000100	222122	$(1, 3, 11)$	$(1, 1, 2, 2, 4, 1)$	$E < 5/4$
010000	212222	$(8, 9, 10)$	$(4, 2, 1, 1, 1, 2)$	$E < 5/4$
100000	122222	$(8, 9, 10)$	$(2, 4, 1, 1, 1, 2)$	$E < 5/4$
000011	222211	$(1, 2, 6)$	$(1, 1, 1, 4, 2, 0)$	$E < 9/8$
000101	222121	$(1, 2, 6)$	$(1, 1, 1, 2, 4, 0)$	$E < 9/8$
001010	221212	$(11, 5, 1)$	$(1, 1, 0, 4, 2, 1)$	$E < 9/8$
001100	221122	$(11, 5, 1)$	$(1, 1, 0, 2, 4, 1)$	$E < 9/8$
010001	212221	$(8, 7, 6)$	$(4, 2, 1, 1, 1, 0)$	$E < 9/8$
100001	122221	$(8, 7, 6)$	$(2, 4, 1, 1, 1, 0)$	$E < 9/8$
011000	211222	$(8, 2, 1)$	$(4, 2, 0, 1, 1, 1)$	$E < 9/8$
101000	121222	$(8, 2, 1)$	$(2, 4, 0, 1, 1, 1)$	$E < 9/8$
001011	221211	$(2, 9, 5)$	$(2, 2, 1, 4, 2, 1)$	$E < 3/2$
001101	221121	$(2, 9, 5)$	$(2, 2, 1, 2, 4, 1)$	$E < 3/2$
011001	211221	$(4, 9, 7)$	$(4, 2, 1, 2, 2, 1)$	$E < 3/2$
101001	121221	$(4, 9, 7)$	$(2, 4, 1, 2, 2, 1)$	$E < 3/2$
010010	212212	$(8, 8, 0)$	$(4, 2, 0, 4, 2, 0)$	$E = 2$
010100	212122	$(8, 8, 0)$	$(4, 2, 0, 2, 4, 0)$	$E = 2$
100010	122212	$(8, 8, 0)$	$(2, 4, 0, 4, 2, 0)$	$E = 2$
100100	122122	$(8, 8, 0)$	$(2, 4, 0, 2, 4, 0)$	$E = 2$
011011	211211	$(4, 10, 6)$	$(4, 2, 0, 4, 2, 0)$	$E = 2$
011101	211121	$(4, 10, 6)$	$(4, 2, 0, 2, 4, 0)$	$E = 2$
101011	121211	$(4, 10, 6)$	$(2, 4, 0, 4, 2, 0)$	$E = 2$
101101	121121	$(4, 10, 6)$	$(2, 4, 0, 2, 4, 0)$	$E = 2$

For a paper-only check of the last column, substitute the integer angles in (16), reduce by the half-angle identity $\tan^2(x/2) = (1-\cos x)/(1+\cos x)$, and use

$$\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}, \quad \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos\frac{\pi}{3} = \frac{1}{2}.$$

Clearing positive denominators and squaring positive sides proves the stated rational comparisons. In each of the eight equality rows the six terms are, up to exchanging members within a doubled pair,

$$(2/3, 1/3, 0, 2/3, 1/3, 0),$$

so equality is exact. The ledger contains $2+2+4+4+4+4+8 = 28$ rows from the old 36-word no-odd–odd slice. Its eight remaining rows are

$$010011,\ 010101,\ 011010,\ 011100, \quad 100011,\ 100101,\ 101010,\ 101100;$$

these are the exceptional automorphism orbit (four rows in class 110 and four in class 111) and are closed by the long-path certificates and structural deletion in the proof of Lemma 5.2. The other 28 realizable parity words, each having an odd–odd doubled side, are closed by the strict twelve-state DNN ledger in that proof. Thus the complete census is $28 + 8 + 28 = 64$, not 36.

B Complete doubled-triangle rational ledger

We similarly expand the twelve-orbit table used in Lemma 5.3. For a doubled side of type $S \in \{EE, EO, OO\}$ and rational parameter t, put

$$r_{EE}(t) = r_{EO}(t) = \frac{1-6t^2+t^4}{(1+t^2)^2}, \qquad e_{EE}(t) = 4t^2,$$

$$e_{EO}(t) = \left(\frac{1-t^2}{2t}\right)^2 + 2t^2,$$

$$q(t) = \frac{3t-t^3}{1-3t^2}, \qquad r_{OO}(t) = -\frac{1-q(t)^2}{1+q(t)^2}, \qquad e_{OO}(t) = q(t)^2 + 3t^2.$$

For the connector define

$$r_E(t) = \frac{1-6t^2+t^4}{(1+t^2)^2}, \quad e_E(t) = 2t^2, \qquad r_O(t) = \frac{t^2-1}{t^2+1}, \quad e_O(t) = t^2.$$

Given (t_1, t_2, t_c), the table uses

$$R = \begin{pmatrix} 1 & r_1 & r_2 \\ r_1 & 1 & r_c \\ r_2 & r_c & 1 \end{pmatrix}, \qquad E = e_{S_1}(t_1) + e_{S_2}(t_2) + e_P(t_c),$$

where P is the connector parity. These multiple-angle formulas reconstruct every parameterized row using rational arithmetic only. The three boundary rows use their displayed correlations directly.

Table 5: Complete doubled-triangle rational certificate table.
The orbit sizes total 32 labelled physical rows.

types	c	size	(t_1, t_2, t_c)	(r_1, r_2, r_c)	E	$\det R$
EE, EE	E	1	$(0, 0, 0)$	$(1, 1, 1)$	0	0
EE, EE	O	1	$(1/3, 1/3, 1/2)$	$(7/25, 7/25, -3/5)$	41/36	1216/3125
EE, EO	E	4	$(0, 1/2, 1/2)$	$(1, -7/25, -7/25)$	25/16	0
EE, EO	O	4	$(1/4, 1/2, 1/3)$	$(161/289, -7/25, -4/5)$	205/144	11527191/52200625
EE, OO	E	2	$(1/4, 1/4, 1/3)$	$(161/289, -495/4913, 7/25)$	2246/1521	8593933244/15085980625
EE, OO	O	2	boundary	$(1, -1, -1)$	0	0
EO, EO	E	4	boundary	$(-1/2, -1/2, 1)$	2	0
EO, EO	O	4	class 111	—	structural	—
EO, OO	E	4	$(1/2, 1/5, 1/5)$	$(-7/25, -828/2197, 119/169)$	83009/48400	1304316959/3016755625
EO, OO	O	4	$(1/2, 1/5, 1/2)$	$(-7/25, -828/2197, -3/5)$	91237/48400	883705599/3016755625
OO, OO	E	1	boundary	$(-1, -1, 1)$	0	0
OO, OO	O	1	$(1/5, 1/5, 1/2)$	$(-828/2197, -828/2197, -3/5)$	16881/12100	22382224/120670225

For each nonstructural row, $|r_1|, |r_2|, |r_c| \le 1$ and the displayed determinant is nonnegative; hence all principal minors of R are nonnegative and $R \succeq 0$. Direct substitution gives the displayed excess fraction. The only less immediate threshold checks are

$$2246 < 2(1521), \quad 83009 < 2(48400), \quad 91237 < 2(48400), \quad 16881 < 2(12100),$$

and every other DNN row visibly has $E \le 2$. The size column sums to 16 rows for each connector parity, or 32 in all; the structural orbit has size four, leaving 28 labelled DNN rows. This supplies all correlations, excesses, determinants, and multiplicities needed to reconstruct the finite part of the doubled-triangle proof from the paper alone.

C Reproducibility, dependencies, and scope

The detailed exact tables summarized above are repository proof objects, not empirical graph censuses. They prove arbitrary path lengths by the monotonic path lemma and arbitrary rooted trees by one-vertex DNN additivity or explicit territory ownership. The source notes are:

`positive-square-energy/tricyclic-general/four-path-dnn-theorem.md`
`positive-square-energy/tricyclic-general/theta-cycle-dnn-reduction.md`
`positive-square-energy/tricyclic-general/theta-cycle-completion.md`
`positive-square-energy/tricyclic-general/doubled-c4-switching-sieve.md`
`positive-square-energy/tricyclic-general/k4-seven-switching-classes.md`
`positive-square-energy/tricyclic-general/k4-all-odd-dnn-cover.md`
`positive-square-energy/tricyclic-general/doubled-triangle-dnn-cover.md`

The finite rational audit checks twelve doubled-triangle type/parity orbits (32 labelled physical rows), partitioned accurately as 28 DNN rows and four structural class-111 rows; it also checks the two exact rational tangent certificates used when a class-111 parallel path is noncanonical. It checks the doubled-4-cycle old slice into 28 DNN rows and an eight-row exceptional orbit split equally between classes 110 and 111, plus the twelve normalized odd–odd Gram records covering all 20 relevant type states and all 28 formerly omitted labelled words. It checks the strict upper bound $229/150 < 2$ for that repair and all 56 physical K_4 rows outside the all-odd switching class. The separate all-odd-class audit checks all eight physical switching rows, the unit/long lower-length rules, and the exact physical Gram for the four three-edge-cut rows. For the physically all-odd row, the finite audit also enumerates all 64 long/unit subsets. The latter split into 42 DNN-audited subsets with at least three long paths, 12 adjacent and three opposite DNN-audited two-long subsets, six one-long structural deletions, and one no-long attached-K_4 packet. Thus 57 subsets are audited by the displayed DNN bounds and seven have structural dispositions. The audit also checks nine four-path representative symbolic records, including the one-unit $e = 0$ endpoint case, and the rational tangent bounds occurring in those records. These records do not constitute a machine proof of the analytic tangent lemma or its all-length coverage. Its canonical payload also commits to all audited doubled-4-cycle class counts, first and failed rows, class-111 long records, endpoint definitions, tangent constants, and the odd–odd ledger, together with the K_4 endpoint and arithmetic constants. Numeric leaves are restricted to integers and rational strings: floats and all other non-integral numeric types are rejected explicitly. The verifier rejects 29 deliberately corrupted certificate payloads, including an omitted odd–odd state and a changed length-three angle. For the 56-row audit it reconstructs every switching invariant and physical sign transport while retaining the physical path parity. Run the audit both normally and with Python assertions disabled:

```
python3 research/tricyclic-finite-rational-certificates-verifier.py
python3 -O research/tricyclic-finite-rational-certificates-verifier.py
python3 positive-square-energy/experiments/k4_all_odd_exact_verify.py
python3 -O positive-square-energy/experiments/k4_all_odd_exact_verify.py
python3 research/doubled-triangle-all-length-certificate.py
python3 -O research/doubled-triangle-all-length-certificate.py
python3 research/tricyclic-gram-obstruction-verifier.py
```

The first two commands must report the certificate digest

```
80cfecc4db26500f7526f635ab36dd7f8bd7aae0f06ab8b89125148e7756e859
```

This canonical payload digest is embedded in the verifier and mirrored in the README manifest. Those two commands must also report 29 rejected hostile mutations. The standalone K_4 audit reports digest `b0c4ed64a12ccabe0a8a38cb9fdb85622d8f7c2d07bdf53f75eb1700a684fa39` and eight rejected mutations. The standalone doubled-triangle audit reports digest `b293eef0d7742da6ecd2c7af358` and three rejected mutations. The final command audits the Gram obstruction. More generally, these verifiers do not prove DNN duality, path elimination or monotonicity, induced superadditivity, the kernel classification, the territory repairs, or the attached-K_4 Sachs argument. They are fail-closed exact audits of only the finite records just enumerated. Build this manuscript from the repository root with

```
bash scripts/build-paper.sh all-tricyclic-graphs
```

Scoped nonclaim. The theorem concerns exactly finite simple connected graphs with $m = n+2$. Some branches above are strict, but DNN certificates may meet their auxiliary boundary. We therefore claim only $s^+(G) \geq n$ globally. We do not claim that all DNN equality cases have been spectrally classified, nor do we invoke edge monotonicity or the false inequality $s^+(G) \geq \mathrm{MaxCut}(G)$.

AI disclosure

This manuscript and its proof-object synthesis were generated by an AI coding and mathematical reasoning system from the cited repository materials. No human authorship, review, or verification is claimed. The result should be assessed from the displayed proof and its exact local dependencies. No external publication was performed as part of this work.

References

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[5] Companion manuscript, *Positive Square Energy of Every Nonunicyclic Cactus*, 2026; `all-nonunicyclic-cacti/paper.tex`.

[6] *The doubled-C_4 kernel: exact switching sieve*, proof note, 2026; `positive-square-energy/tricyclic-general/doubled-c4-switching-sieve.md`.

[7] *The doubled-triangle kernel: complete physical parity cover*, proof note, 2026; `positive-square-energy/tricyclic-general/doubled-triangle-dnn-cover.md`.

[8] *The all-odd K_4-subdivision class*, proof note, 2026; `positive-square-energy/tricyclic-general/k4-all-odd-dnn-cover.md`.

[9] *The seven non-all-odd switching classes of subdivided K_4*, proof note, 2026; `positive-square-energy/tricyclic-general/k4-seven-switching-classes.md`.

[10] Companion manuscript, *Square-energy sign under cycle packing number at most two*, 2026; `packing-two-square-energy/paper.tex`.